

Problem with Perceptrons

1 What is the Problem?

- A **single-layer perceptron** can only draw a **straight line (or plane)** to separate classes.
 - If the data is **linearly separable** (can be split by one line), the perceptron works fine.
 - But if the data is **not linearly separable** → the perceptron fails.
-

2 Example: XOR Problem

- Imagine data points:
 1. Class 1: (0,1), (1,0)
 2. Class 0: (0,0), (1,1)
- If you try to draw a single straight line, you **can't separate** the two classes.
- This is called the **XOR problem** (exclusive OR).

A perceptron **cannot learn XOR**, because it can't make a curved or multi-step boundary.

3 Why Does This Happen?

- The perceptron uses a **linear function** to compute output.
 - The step activation only gives binary decisions.
 - It lacks the ability to combine features in a **non-linear way**.
-

4 Consequences

- Perceptrons **can't learn complex patterns** like:
 1. Circles, spirals, XOR, or overlapping regions.
 2. Any problem that needs multiple decision boundaries.
-

5 Solution

This limitation led to the development of:

- **Multi-Layer Perceptrons (MLPs)**: Add hidden layers with non-linear activations (sigmoid, ReLU, tanh).
- **Deep Neural Networks**: Stack many layers to learn highly complex patterns.

With extra layers, neural networks can approximate **any function** (Universal Approximation Theorem).

Quick Recap

- **Perceptron strength**: Works well for linearly separable data.
-

- **Perceptron weakness:** Fails on non-linear problems (e.g., XOR).
 - **Fix:** Add more layers + non-linear activations → modern deep learning.
-